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Proton structure !

Generating the transverse distributions from first principles

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Résumé

Abstract

Introduction

Idées principales, contexte, intérêt, déroulé

Citations temporaires pour que tout apparaisse dans la bibliographie : [5], [3], [4], [2], [1]

Chapter 1

[Elements of QFT]

Quelques mots sur les concepts de base de la QFT (champs, particules, méthode perturbative, règles de Feynman, algèbre, modèle standard). Variation d'α_s (explications sur la renormalisation en annexe). Confinement.

Chapter 2

[Proton structure]

Passage sur les DIS. DGLAP. Bases du parton branching.

Chapter 3

A simplified parton evolution

In this section, we will show how to numerically compute PDFs and obtain transverse momentum distributions with a parton branching method, with a very simplified framework, taken from the exercises adjoining [3]. Even if this implementation is far less sophisticated than the one this thesis is focused on, it already holds many of its key points, and will allow us, by fiddling with its parameters, to investigate its behaviour and reveal points of interest to keep in mind for the full implementation.
Note : *This is the reason why completing the exercises from [3] was the first step of my assignment. The parton evolution presented here is a slight variation of exercise 3 of these lectures. I used [3] and [4] as a guide to complete it.*

3.1 Implementation

We want to evolve a starting gluon density $f(x, \mu_0^2) = 3(1-x)^5/x$ from scale $\mu_0^2 = 1\,\text{GeV}^2$ to scale $\mu_f^2 = 10\,000\,\text{GeV}^2$. We will also generate their transverse momentum k_t during the evolution, and assuming a starting distribution with zero mean and $\sigma = 0.7\,\text{GeV}$. We will use the splitting function

$$P_{gg}(z) = 6\left(\frac{1}{z} + \frac{1}{1-z}\right) \tag{3.1}$$

with cut-offs $z_{min} = \epsilon$, $z_{max} = 1 - \epsilon$ with $\epsilon = 0.1$. As a last simplification, we will take the strong coupling to be constant : $\alpha_s = 0.1$.
We will also redo the same exercise, with a quark-quark splitting

$$P_{qq}(z) = \frac{4}{3}\frac{1+z^2}{1-z} \tag{3.2}$$

and assuming the same starting distribution (which could pass as an approximation of a sea quark density, but not a valence quark density).

In order to evaluate the density at the final scale, $f(x, \mu_f^2)$, we will generate 1 000 000 events from the starting density and evolve them by simulating their possible branching according to the known probability densities. The evolved events will then be distributed following $f(x, \mu_f^2)$. During the evolution of each event, we will also generate the k_t of the emitted gluon, and compute the change of k_t of the emitter, to obtain a k_t distribution at the final scale.

For each event, we must perform the following steps :

1. **Generate the starting momentum fraction x_0 and transverse momentum $k_{t,0}$:**

   Here, we generate x_0 with the importance sampling method, using the approximate function

$g(x) = \frac{1}{x \log(x_{max}/x_{min})}$, with $x_{min} = 0.000\,01$ and $x_{max} = 1$.
Then, drawing a uniform random number $R_{x_0} \in [0, 1]$, we get x_0 and its weight w_{x_0} as

$$\boxed{x_0 = x_{min} \left(\frac{x_{max}}{x_{min}}\right)^{R_{x_0}}} \tag{3.3}$$

$$\boxed{w_{x_0} = 3\,(1 - x_0)^5 \, \log(x_{min}/x_{max})} \tag{3.4}$$

We generate $k_{t,0}$ using the central limit theorem, by summing 12 uniform random numbers $R_i \in [0, 1]$, with then

$$\boxed{k_{t,0} = 0.7 \left(\sum_{i=1}^{12} R_i - 6\right)} \tag{3.5}$$

These will indeed distribute following a Gaussian distribution of mean 0 and standard deviation 0.7. We also need the two components $k_{x,0}$ and $k_{y,0}$ for later use, so we draw a uniformly distributed azimuthal angle $\phi \in [0, 2\pi]$ to initialise them at $k_{x,0} = k_{t,0} \cos(\phi)$, $k_{y,0} = k_{t,0} \sin(\phi)$.

2. **Generate the next branching scale μ_i^2**

We generate the next scale μ_i^2 with respect to the Sudakov form factor

$$\Delta_S(\mu_{i-1}^2, \mu_i^2) = \exp\left(-\int_{\mu_{i-1}^2}^{\mu_i^2} \frac{d\mu^2}{\mu^2} \int_{z_{min}}^{z_{max}} dz \frac{\alpha_s}{2\pi} P_{gg}(z)\right) \tag{3.6}$$

by drawing a uniformly distributed random number $R_{\mu_i^2} \in [0, 1]$ and inverting the relation $R_{\mu_i^2} = \Delta_S(\mu_{i-1}^2, \mu_i^2)$. In our case, it is easy since the only scale dependence is in the $1/\mu^2$ term, as we are approximating α_s to be constant. Thus we have

$$\begin{aligned}
&\ln(R_{\mu_i^2}) = -\int_{\mu_{i-1}^2}^{\mu_i^2} \frac{d\mu^2}{\mu^2} \int_{z_{min}}^{z_{max}} dz \frac{\alpha_s}{2\pi} 6 \left(\frac{1}{z} + \frac{1}{1-z}\right) \\
\Leftrightarrow \quad &\ln(R_{\mu_i^2}) = -\frac{\alpha_s}{2\pi} \ln\left(\frac{\mu_i^2}{\mu_{i-1}^2}\right) 6 \left[\ln\left(\frac{z_{max}}{1 - z_{max}}\right) - \ln\left(\frac{z_{min}}{1 - z_{min}}\right)\right] \\
\Leftrightarrow \quad &\ln\left(\frac{\mu_i^2}{\mu_{i-1}^2}\right) = -\frac{\pi}{3\alpha_s} \ln\left(\frac{z_{max}(1 - z_{min})}{(1 - z_{max}) z_{min}}\right)^{-1} \ln(R_{\mu_i^2}) \\
\Leftrightarrow \quad &\boxed{\mu_i^2 = \mu_{i-1}^2 \, R_{\mu_i^2}^{-\frac{\pi}{3\alpha_s} \ln\left(\frac{z_{max}(1-z_{min})}{(1-z_{max})z_{min}}\right)^{-1}}}
\end{aligned} \tag{3.7}$$

With the quark-quark splitting function $P_{qq}(z)$ from Equation 3.2, the exponent becomes

$$-\frac{3\pi}{2\alpha_s} \left(\frac{z_{min}^2 - z_{max}^2}{2} + (z_{min} - z_{max}) + 2 \ln\left(\frac{1 - z_{min}}{1 - z_{max}}\right)\right)^{-1} \tag{3.8}$$

We then compare the new scale with the final scale : if $\mu_i^2 > \mu_f^2$, this means that no branching occurred up to the final scale, and we keep the last x_i and $k_{t,i}$ generated. If not, then a branching occurred at this scale, and we carry on with the following steps to generate its effect.

3. **Generate the splitting variable z_i and the new $k_{t,i}$**

The splitting variable is generated in a similar way, using the normalised integrated splitting function as the cumulative function. So, with a uniformly generated random number $R_{z,i} \in [0,1]$, we have

$$\begin{aligned}
R_{z,i} &= \frac{\int_{z_{min}}^{z_i} dz\, P_{gg}(z)}{\int_{z_{min}}^{z_{max}} dz\, P_{gg}(z)} \\
\Leftrightarrow \quad R_{z,i} &= \frac{6\left(\ln\left(\frac{z_i}{1-z_i}\right) - \ln\left(\frac{z_{min}}{1-z_{min}}\right)\right)}{6\left(\ln\left(\frac{z_{max}}{1-z_{max}}\right) - \ln\left(\frac{z_{min}}{1-z_{min}}\right)\right)} \\
\text{so with} \quad & \boxed{A = R_{z,i} \ln\left(\frac{z_{max}}{1-z_{max}}\right) + (1-R_{z,i}) \ln\left(\frac{z_{min}}{1-z_{min}}\right)} \\
\Leftrightarrow \quad \frac{z_i}{1-z_i} &= e^A \\
\Leftrightarrow \quad & \boxed{z_i = \frac{e^A}{1+e^A}}
\end{aligned} \tag{3.9}$$

For the quark-quark splitting of Equation 3.2, inverting the relation is more tricky. We prefer to use importance sampling again, using as an approximate function $g(z) = \frac{1}{(1-z)\ln(1-z_{min}/1-z_{max})}$ with thus

$$z_i = 1 - (1-z_{min})\left(\frac{1-z_{max}}{1-z_{min}}\right)^{R_{z_i}} \tag{3.10}$$

with weight

$$w_{z_i} = \frac{4}{3}(1+z_i^2)\ln(1-z_{min}/1-z_{max}) \tag{3.11}$$

We can then update the momentum fraction by multiplying the previous one with the new splitting variable :

$$x_i = z_i\, x_{i-1} \tag{3.12}$$

In the quark case, we also need to multiply the weight : $w_{x_i} = w_{z_i}\, w_{x_{i-1}}$ while in the gluon case, the weight stays at its initial value from Equation 3.4.

We can also generate the new $k_{t,i}$. Actually, we generate the k_t of the emitted particle, then calculate the resulting $k_{t,i}$ of the emitter. The choice of expression for the k_t is not set in stone, we have to make a heuristic choice. Here, we will investigate two options : *transverse ordering* and *angular ordering*. The former prescribe

$$k_t = \mu_i \tag{3.13}$$

Where the natural scale for the k_t is then taken to be the current energy scale. The latter uses

$$k_t = \mu_i (1 - z_i) \tag{3.14}$$

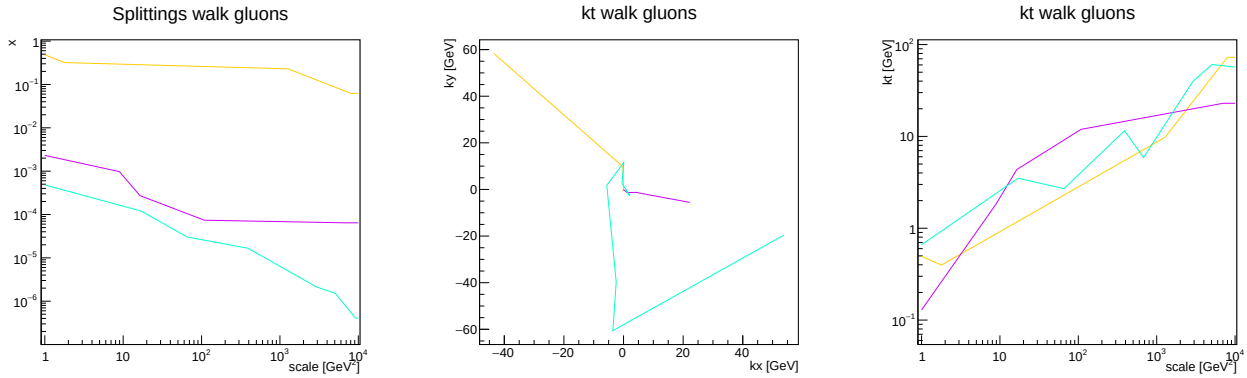
Which rather associate the energy scale with the angle of the momentum of the emitted particle, with respect to the one of the emitter. The discussion on how these are obtained is deferred to the next chapter. In the meantime, we will just use them as-is to investigate the resulting distributions.
In the following analysis, we will try angular ordering first, and transverse ordering second.

Generating a uniformly distributed azimuthal angle $\phi \in [0, 2\pi]$, we can finally evaluate

$$\boxed{k_{x_i} = k_{x_{i-1}} - k_t \cos\phi \quad ; \quad k_{y_i} = k_{y_{i-1}} - k_t \sin\phi \quad ; \quad k_{t_i} = \sqrt{k_{x_i}^2 + k_{y_i}^2}} \tag{3.15}$$

4. **Jump back to 2. until the end of the loop**

Going through these steps looks like this (with angular ordering for the k_t generation) :



(a) sample gluon evolution (b) sample gluon k_t generation (c) sample gluon k_t evolution

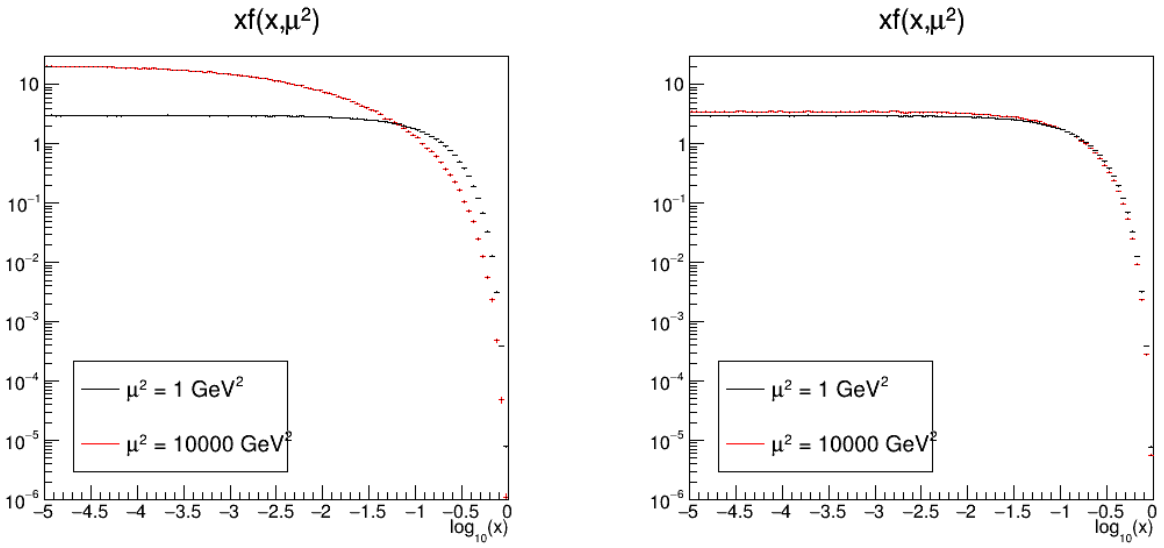
Figure 3.1: Evolution process for a sample of three gluon.

On Fig. 3.12a, we can see the momentum fraction being reduced as splittings occur through the scale evolution. Fig. 3.12b shows the evolution of the x and y components of the transverse momentum as branchings occur, and Fig. 3.1c plots the k_t with respect to the scale.

To get a significant sample, we perform this for 1 000 000 events, and plots the resulting densities in histograms. Note that in order to obtain histograms for the weighted density $x f(x, \mu^2)$, we need to multiply the weight for each x by x itself. There is one last thing to take into account to get the correct normalisation : this method evolves as many partons as that are generated for the initial distribution, but their momentum fraction only lowers. This seems in conflict with the fact that the integral of the momentum density must of course be conserved. It is so because the momentum fraction is not lost, but spread between more partons. To account for this, we must simply normalise the evolved densities to the same momentum integral as the initial ones. We can also note that only the $f(x, \mu^2)$ integrals are conserved through evolution, not the $x f(x, \mu^2)$ nor the k_t ones. (The former because of the shift in the distribution of x, the latter because of the increase of the number of partons : more high-k_t partons does not automatically indicate less low-k_t ones).

3.2 Results

Here are the resulting histograms :



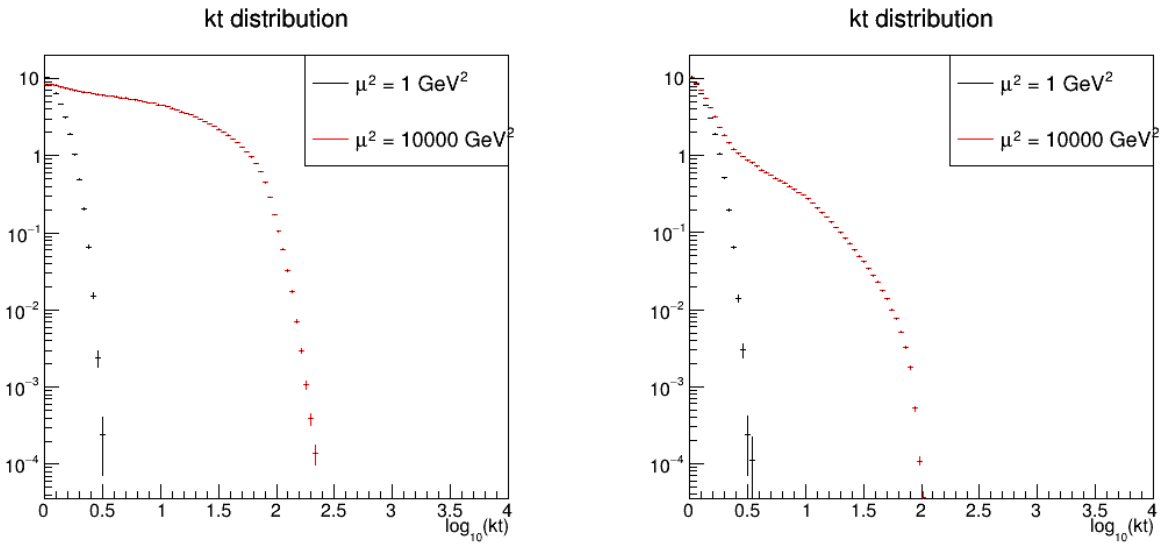
(a) gluon density

(b) quark density

Figure 3.2: Evolution of the momentum weighted parton density.

In the gluon case, the migration to small momentum fractions x is plain to see, while in the quark case it is much less pronounced. Seeing the latter, we might fear an inadequacy in the parameters used, perhaps in the cut-off. We will see that the cut-off does play a role, but is also counterbalanced. This result is actually robust, in a sense that we will explore a bit later. The difference between the results for the gluon and the quark densities really come from the behaviour of their splitting functions.

Here are the results for the transverse momentum k_t (with angular ordering) :



(a) gluon distribution

(b) quark distribution

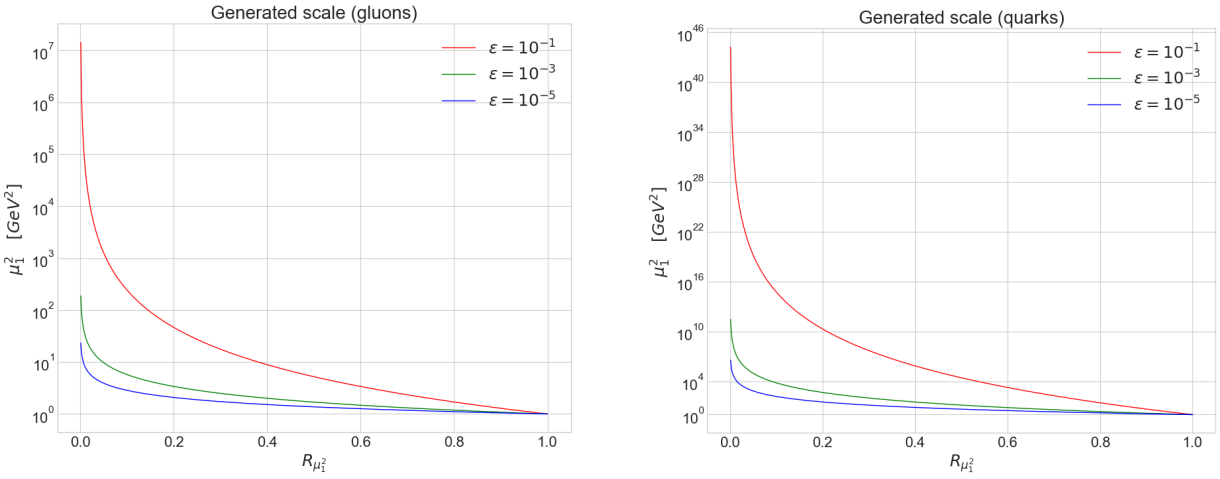
Figure 3.3: Evolution of the transverse momentum distribution.

We can notice that the distributions decrease sharply because of the scale limitation, although it does exceed it in the gluon case, likely because of the accumulated steps. The quark distribution being notably lower might put in conjunction with the less pronounced evolution, but the abrupt stop at the final scale might indicate that

hardly more than one branching ever occur for a single event.

This justifies an investigation on the dependence of these results on, say, the cut-off ϵ, which gives $z_{min} = \epsilon$, $z_{max} = 1 - \epsilon$ and thus control the integrals on the splitting function.

Let us take a look at the ϵ dependence for the scale generation :

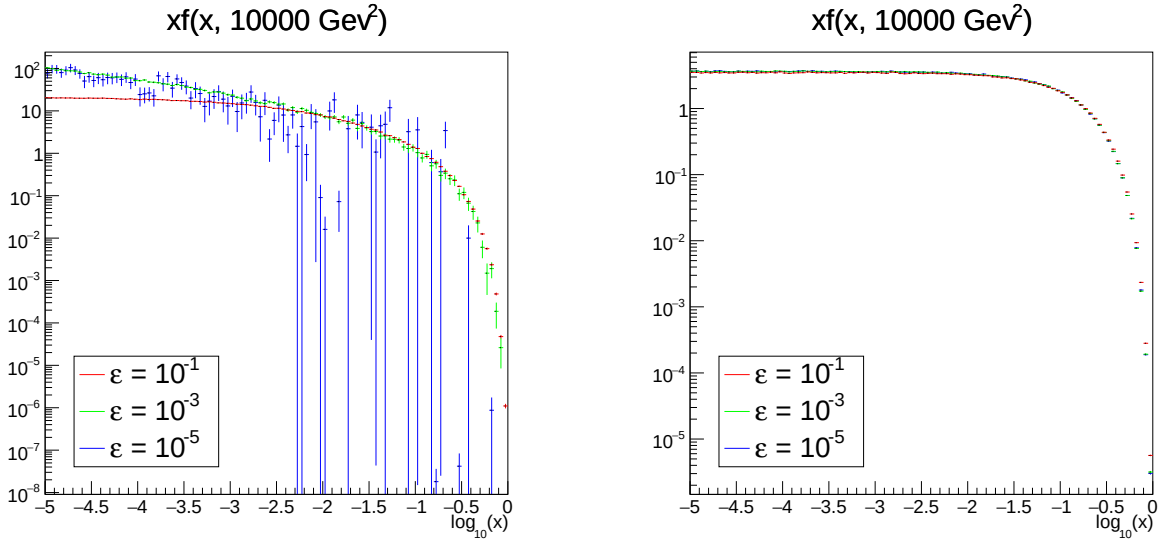


(a) gluon case (b) quark case

Figure 3.4: First generated scale μ_1^2 with respect to the random number $R_{\mu_1^2}$, depending on ϵ.

Let us recall that the evolution is stopped when $\mu_i^2 > \mu_f^2 \equiv 10^4\,\text{GeV}^2$. Let us also recall that from Equation 3.7, with a different initial scale $\mu_0^2 \neq 1$ we would have got the same graph, with μ_1^2 multiplied by μ_0^2.
We can see that decreasing ϵ decrease the probability of generating large scales (a smaller range of $R_{\mu_1^2}$ would produce them), which overall means that there will be more branching before the final scale is reached. We can also note that the scales generated in the quark case are often very large, especially for $\epsilon = 0.1$. In this latter case, for more than half the events, no branching would occur at all (as for $R_{\mu_1^2} \in (0, 0.54)$, we have $\mu_1^2 > \mu_f^2$).

Let us have a look at how ϵ affects the density plots :

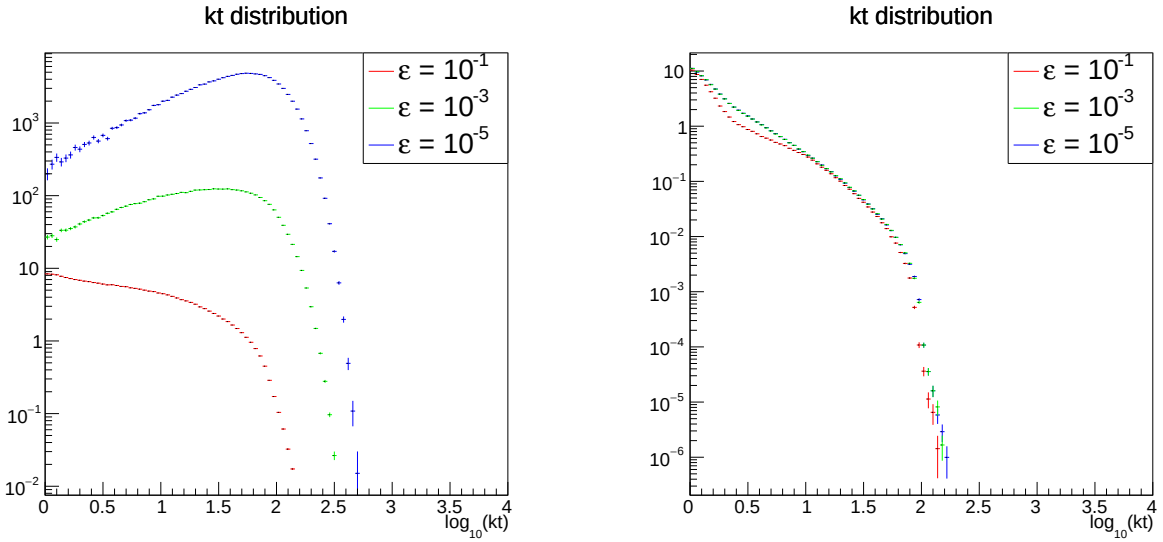


(a) gluon density (b) quark density

Figure 3.5: Evolved momentum weighted parton density, depending on ϵ.

The gluon density is strongly affected : decreasing ϵ strongly increase the density for small momentum fractions.

Note : *The $\epsilon = 10^{-5}$ case is messy because of scarcely filled bins : on 1 000 000 events generated, only 76 end up in the right half of the plot (the majority is actually in the underflow).*
Perhaps surprisingly, the quark density seems mostly unaffected by the variation of ϵ. We need to look at the k_t distribution to find a difference :

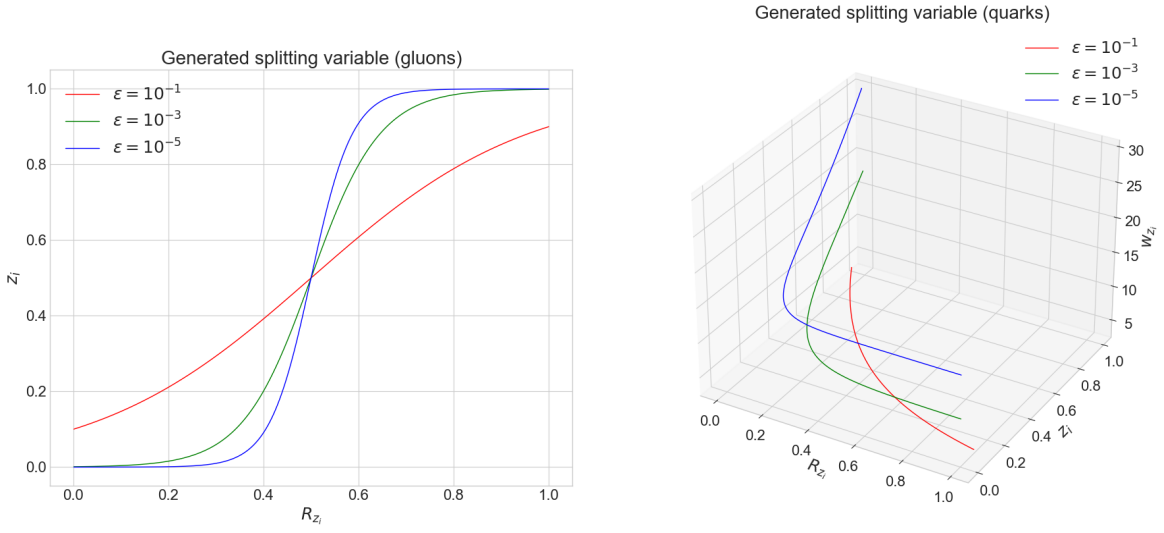


(a) gluon distribution

(b) quark distribution

Figure 3.6: Evolved transverse momentum distribution, depending on ϵ.

The gluon density is still strongly affected, while for the quark density, the $\epsilon = 10^{-3}$ and $\epsilon = 10^{-5}$ straighten up a bit from the $\epsilon = 10^{-1}$ case, but match with each other. We can suppose that with ϵ to large, there weren't quite enough branchings generated to get a statistically sound k_t distribution, while the quark distribution otherwise don't seem to depend on it. This is not the case for the gluon distributions. To understand this better, we need to take a look at the ϵ dependence of the splitting variable generation :



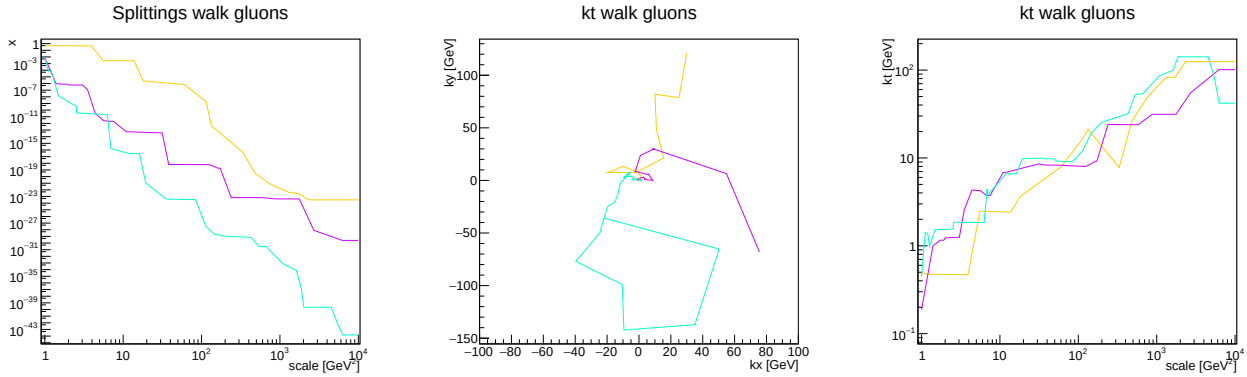
(a) gluon case

(b) quark case, showing the weights

Figure 3.7: Generated splitting variable z_i with respect to the random number R_{z_i}, depending on ϵ.

Since we used the importance sampling method to generate the quark splitting, we need to show the corresponding weights to accurately display the importance of each value in the results. We can notice that for the quark case, reducing ϵ increase the importance of large z_i values. Thus events where the splittings doesn't change the momentum fraction much (z close to 1) weight more in the histogram, while, as seen in Fig. 3.4,

more branchings occur and produce more splittings, reducing x more. Apparently, these two opposite effects cancel each other in the quark case, giving ϵ-independant results (provided ϵ is not *too* large). This is quite fortunate, since the cut-off is unphysical.
We are not so lucky with the gluon density, but we can see why in Fig. 3.7a : decreasing ϵ makes z_i close to 1 more probable, but also z_i close to 0. This latter effect explain the strong migration towards very small momentum fractions in Fig. 3.5a. We also generate more splittings, and given the formula for the k_t used in Equation 3.14, we make more large jumps in k_t, which explains the large density for large k_t in Fig. 3.13a.
Compared to Fig. 3.12, we now get more events like these :

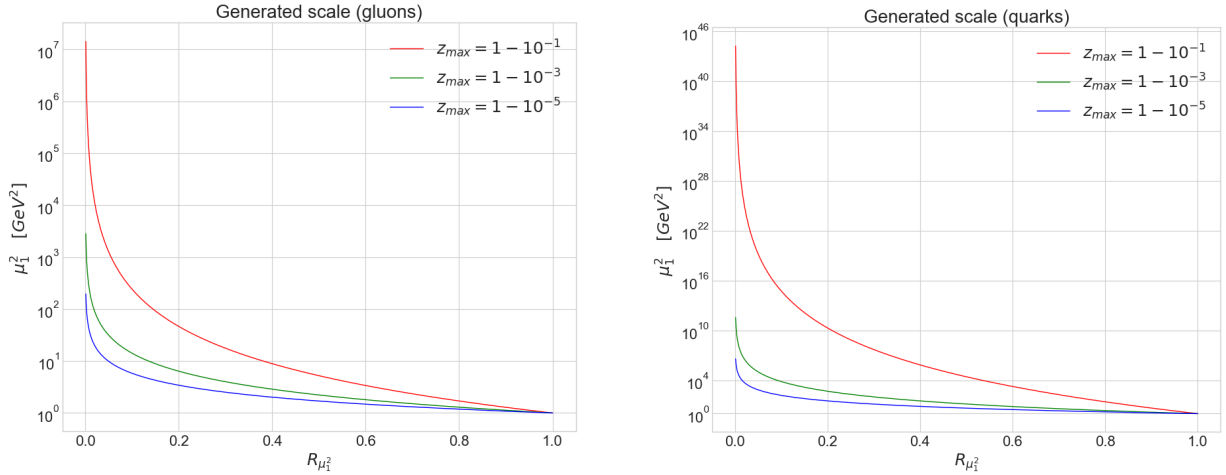


(a) $\epsilon = 10^{-5}$ gluon evolution (b) $\epsilon = 10^{-5}$ gluon k_t generation (c) $\epsilon = 10^{-5}$ gluon k_t evolution

Figure 3.8: Evolution process for a sample of three gluon, with $\epsilon = 10^{-5}$.

These events exemplify well the behaviour discussed : we can see on Fig. 3.8a a lot of splitting who sharply decrease the momentum fraction, and on the remaining plots, the tendency to build a large k_t.

We can then ask ourselves how would this turn out if we fixed z_{min}, in hope to avoid this small z_i issue. With $z_{min} = 0.1$ fixed, the variable generation would then behave this way :

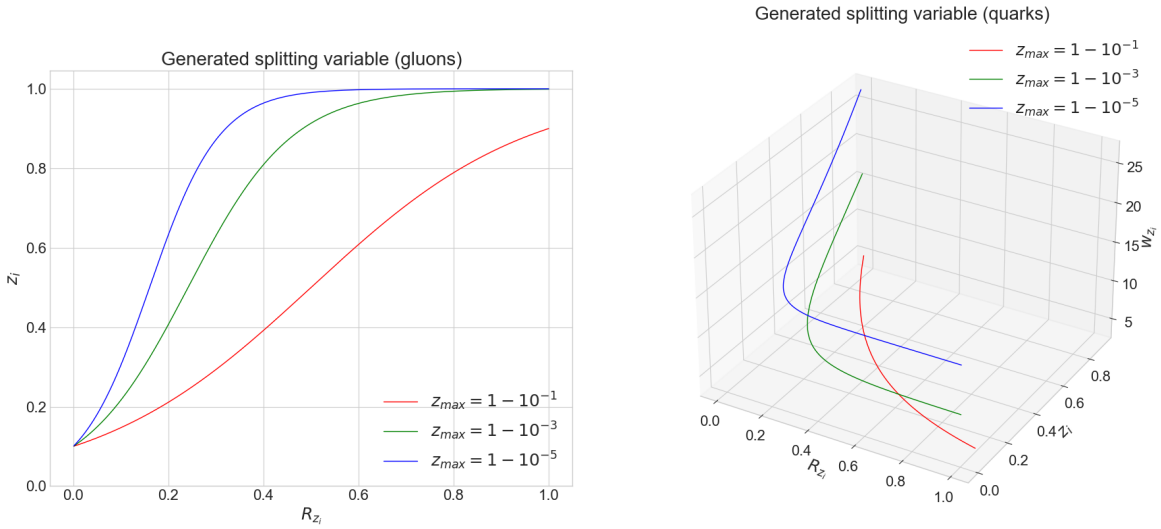


(a) gluon case (b) quark case

Figure 3.9: First generated scale μ_1^2 with respect to $R_{\mu_1^2}$, fixed z_{min}, variable z_{max}.

The generation of the scale is noticeably changed only for the gluon case. This is natural, as the splitting function for the gluon in Equation 3.1 diverge as $z \to 0$, while it is not the case for the quark in Equation 3.2. By fixing z_{min}, we avoid this divergent contribution. So now we have more chances to generate bigger scales, compared to the unfixed z_{min} scales, so we shouldn't get as many splittings per events as we could see in Fig. 3.8a. (To be clear, approaching z_{max} of 1 still increases the amount of splittings, just not as much as when we approached z_{min} from 0 too.)

The change for gluons is especially striking for the splitting variable generation :

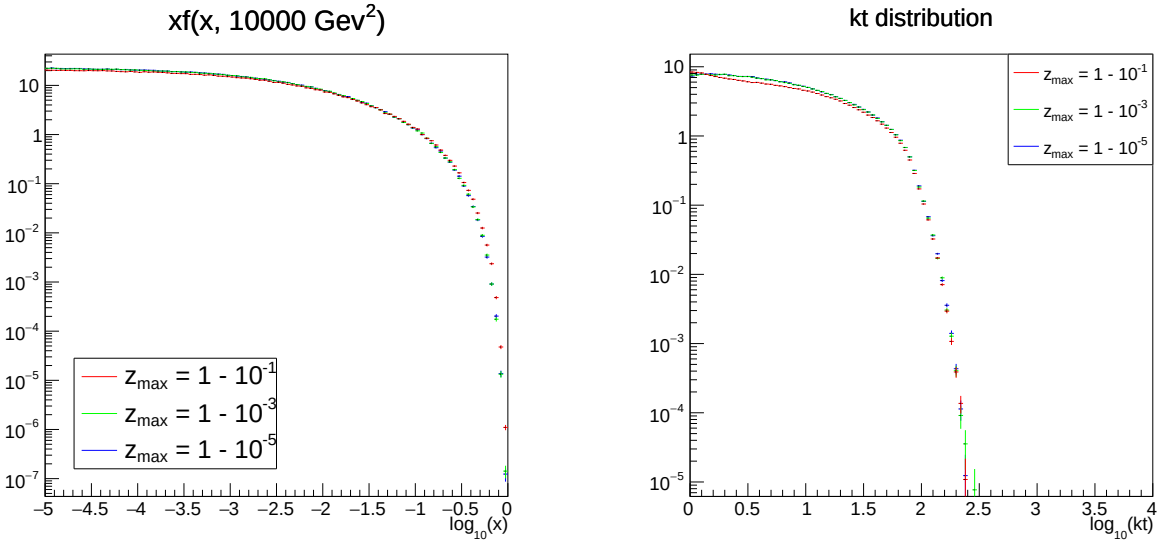


(a) gluon case

(b) quark case, showing the weights

Figure 3.10: Generated splitting variable z_i with respect to R_{z_i}, fixed z_{min}, variable z_{max}.

Now, we can't generate z values arbitrarily close to zero, and small values are becoming less likely as we take z_{max} closer to 1. The splittings will then be less impactful.
For the quarks, the biggest change is in the weight, but it is proportional for each case. Fixing $z_{min} = 0.1$ doesn't change much the results for the quarks, while it very much do for the gluons :



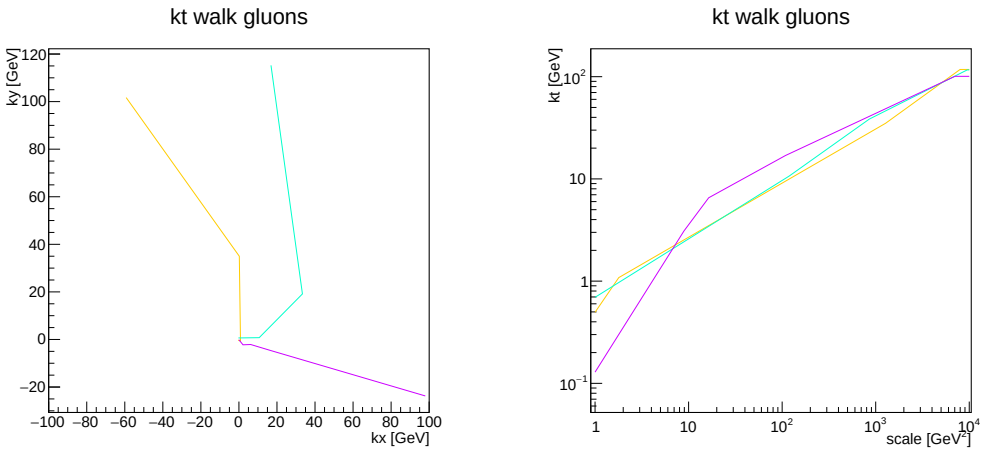
(a) momentum weighted density

(b) transverse momentum distribution

Figure 3.11: Evolved gluon densities, fixed z_{min}, variable z_{max}.

We finally get stable results. We saw here that we can hope to get consistent results independently of the arbitrary choice of z_{max} (provided that it is reasonably close to 1). This is because bringing z_{max} closer to 1 increase the amount of splittings generated, but also increase the probability to generate the splitting variable z_i closer to 1, thus making the splittings less impactful, for both the x and the k_t distribution.

Now we can investigate what happens when we choose transverse ordering (see Equation 3.13) for the k_t generation. It doesn't affect the x density, but it should of course change the k_t densities. Firstly, here is an example of gluon event generation, with $z_{min} = 0.1$ and $z_{max} = 0.9$:

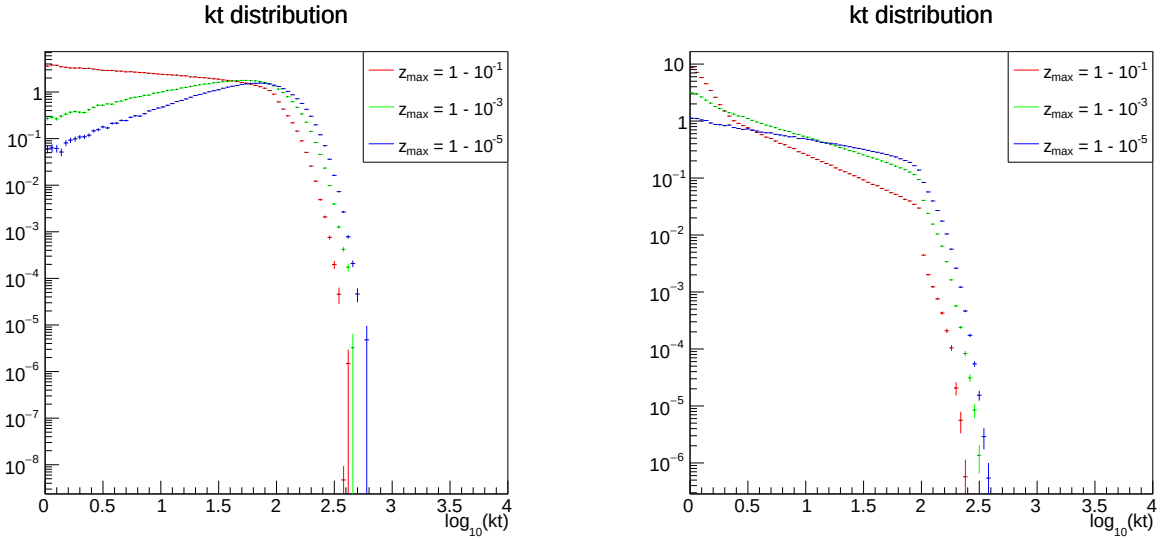


(a) sample gluon k_t generation

(b) sample gluon k_t evolution

Figure 3.12: k_t evolution with transverse ordering, for a sample of three gluons, $z_{max} = 0.9$.

Now successive steps always do a larger k_t jump. But we can already anticipate an issue here. The amount of splitting is sensitive to z_{max}, and so is the z_i generation, but the k_t generation is only sensitive to the former, since Equation 3.13 does not depend on z_i. We can expect to lose z_{max} independence, and this is indeed the case :



(a) gluon distribution

(b) quark distribution

Figure 3.13: Evolved k_t distribution, transverse ordering, variable z_{max}

Angular ordering is a better choice, because it correctly compensates its sensitivity to the amount of splittings with its z_i dependence. We will see in the next chapter another physical argument that could have helped guess its suitability.

Of course, the model used here is too simple to yield physically accurate results. It is nonetheless interesting to see how far (or how close) we can get. Here is a comparison of our toy model with PDFs produced with QCDNUM, both starting with the following realistic parton distributions

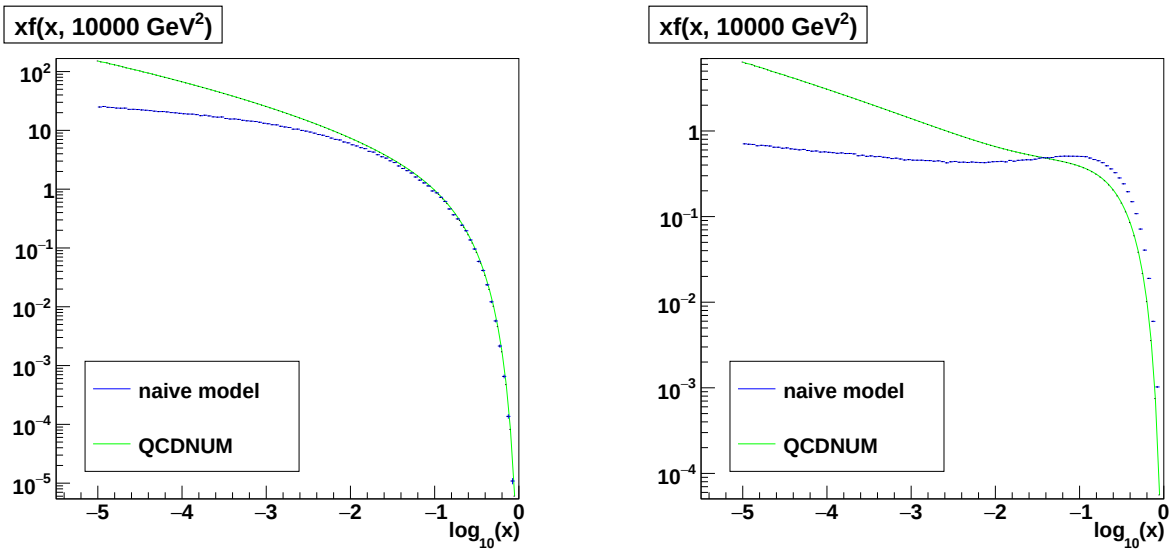
$$\text{gluons :} \quad x f(x, \mu_0^2) = 1.7\, x^{-0.1}(1-x)^5 \tag{3.16}$$

$$\text{down (valence) :} \quad x f(x, \mu_0^2) = 3.064320\, x^{0.8}(1-x)^4 \tag{3.17}$$

$$\text{down (sea)}: \quad x f(x, \mu_0^2) = 0.1939875\, x^{-0.1}(1-x)^6 \tag{3.18}$$

$$\tag{3.19}$$

evolved from $\mu_0^2 = 2\,\text{GeV}^2$ to $\mu_f^2 = 10^4\,\text{GeV}^2$, and where for our "naive" model we took $z_{min} = 0.1$, $z_{max} = 1-10^{-5}$:



(a) gluon density

(b) down quark (valence + sea) density

Figure 3.14: Comparison of momentum weighted parton densities.

We can see that our naive model does not evolve the distributions as far as QCDNUM does. This is not a surprise. First, we took $z_{min} = 0.1$, which is quite high, and so strongly reduces the overall impact of the splittings. We had to restrict ourselves to this here, since we saw at Fig. 3.13a that lowering z_{min} lead to bad k_t distributions. A more accurate model should allow us to use a considerably lower z_{min}. Second, we only do one kind of splitting (gluon-gluon or down-down). But several other forms of splittings are allowed, notably gluon-quark, next-to-leading-order splittings with loops, splittings with other quarks flavours, and so on. Restricting us to one single kind of splitting of course limits the magnitude of the evolution. And finally, we only took leading order splitting functions, and we did not evolve the coupling α_s, which would also affect the splitting process. (We can't say what changes with the higher-order splitting functions without doing the same analysis as with Fig. 3.7, something we leave for a later chapter. We can guess that a running α_s will have a higher value for low scales that the ones used here, thus increasing the probability for splittings to occur.)
We can note on Fig. 3.14a that our naive model is still not so bad for the gluon density at high x values. This is probably because the splittings involving gluons are dominant, and so gluon-gluon splitting only would be valid at high x, while omitting quark-gluons splittings in Fig. 3.14b would be too much of an overlook.

3.3 Summary and perspectives

Résumer les étapes, expliquer ce qui changera dans l'application complète. Rôles différents de z_{min} et z_{max}, trouver une bonne justification pour la valeur de z_{min}.

Chapter 4

[Study of the DGLAP evolution equation]

Sum rules, reformulation, etc...

Chapter 5

À déterminer+

Fonctionnement du code...

Conclusion

Annexes

Annex A

IA use

Bibliography

Textbooks and lecture notes

[1] R. K. Ellis, W. J. Stirling and B. R. Webber. *QCD and Collider Physics*. Cambridge Monographs on Particle Physics, Nuclear Physics and Cosmology. Cambridge University Press, 1996.
Textbook on phenomenological QCD, and topics related to its study at high-energy colliders.

[2] Laurent Favart. "Physique des interactions fortes". Lecture notes from the ULB. 2024-2025.
The notes of the lecture on phenomenological QCD I attended as a graduate student.

[3] Hannes Jung. "QCD and Monte Carlo". Lecture notes from the 2016 Cracow winter course. 2016. URL: https://www.desy.de/~jung/qcd_and_mc_2016/QCD_and_MC_2016_%28Cracow%29.html.
Lecture notes on phenomenological QCD and Monte Carlo methods.

Thesis

[4] Aleksandra Lelek. "Determination of TMD parton densities from HERA data and application to pp processes". PhD thesis. Universität Hamburg, 2018.
On the developpement of the PB-TMD method.

Articles

[5] F. Hautmann et al. "Collinear and TMD quark and gluon densities from parton branching solution of QCD evolution equations". In: *Journal of High Energy Physics* 2018.1 (Jan. 2018). ISSN: 1029-8479. DOI: 10.1007/jhep01(2018)070. URL: http://dx.doi.org/10.1007/JHEP01(2018)070.
On the application of the PB-TMD method at NLO to fit on HERA data.